

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech Winter 2025 - 26 Examination

Semester: 7
Subject Code: 303105289
Subject Name: Big Data on Cloud

Date: 19-12-2025
Time: 2:00pm to 4:30pm
Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1 is compulsory**, From Section B, **Q.1 is compulsory**.
3. Figures to the right indicate full marks.
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION-A		Marks	CO	BT
Q.1	Answer the following questions.			
	A. 1. Define Cloud Security with real life examples. 2. Define Change Data Capture (CDC). 3. What is Cloud Infrastructure Management?	[06]	1	1
	B. 1. Define Feature Extraction. Give real life examples. 2. What is Streaming Data Clustering with real life examples? 3. What is Collaborative Filtering? Give real life examples.	[06]	1	1
Q.2	A. What is Big Data? Explain the characteristics of Big Data with brief notes.	[04]	2	2
	B. Explain how the Apriori Algorithm is applied in Market Basket Analysis.	[05]	3	3
	OR			
	B. Draw a neat diagram and explain the Distributed Data Processing in detail.	[05]	3	3
Q.3	A. What is Big Data Analytics? Explain its applications in real-world scenarios.	[04]	2	2
	B. Explain how PPML techniques protect sensitive data in ML models.	[05]	3	3
	OR			
	B. Compare Supervised vs. Unsupervised Learning in detail with real -world applications.	[05]	3	4

SECTION-B				
Q.1	Answer the following questions.			
	A. 1. What is Spark Streaming? 2. Give two advantages of MongoDB over traditional RDBMS. 3. What is Data Frame?	[06]	2,5	1
	B. 1. What is MongoDB and give features of MongoDB? 2. List down the tools of real time data processing. 3. What is RDD?	[06]	2,5	1
Q.2	A. Differentiate between real-time processing and batch processing.	[04]	2	4
	B. Explain how Spark performs in-memory computation and why it is faster than Hadoop MapReduce.	[05]	2	3
	OR			
	B. Explain what is NoSQL database and discuss its main features and advantages. Give difference between RDBMS and NoSQL.	[05]	5	4
Q.3	A. Explain the categories of NoSQL databases in detail with examples.	[04]	5	2
	B. What is Data Lineage explain in detail? Explain why it is important in big data processing. Give benefits and key components of data lineage.	[05]	3	2
	OR			
	B. “Fraud detection requires real-time processing, whereas payroll systems use batch processing.” – Justify this statement.	[05]	3	4